

Practice Final Test (Version 2)

Name
Date
Course

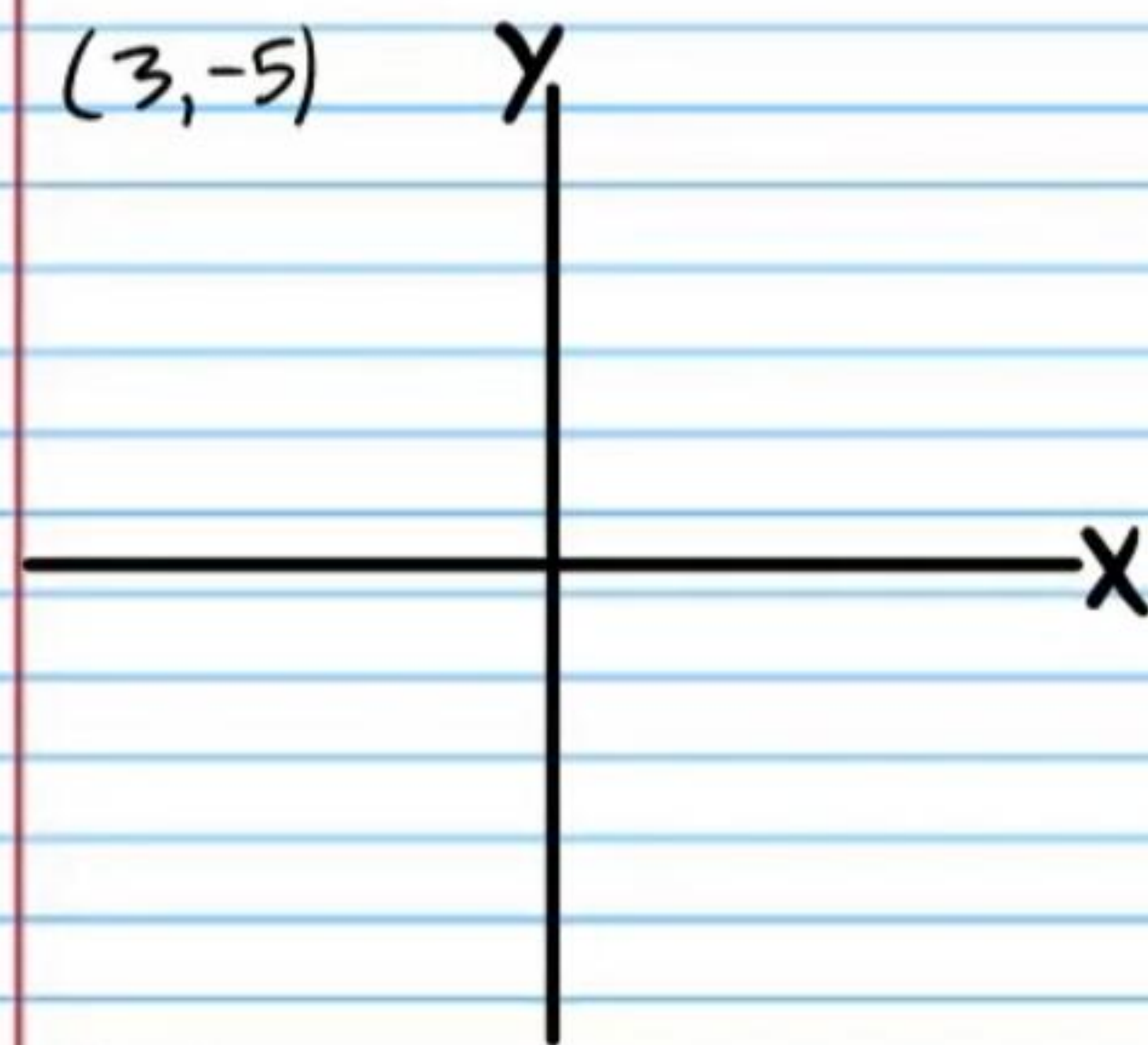
- I) This is a cumulative exam.
- II) Write your name, the date, and the course title (i.e. Trigonometry).
- III) Show all work.
- IV) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- V) You will need a calculator for this test. When π is required, use the π button on your calculator. Do not use 3.14.
- VI) Each problem is worth eight points for a total of 200 points.

① Find the value of the expression below without a calculator.

$$\operatorname{arccsc}\left(-\frac{2}{\sqrt{2}}\right)$$

② The coordinates of a point are given below. If this point lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

(3, -5)



i) $\sin \theta$

ii) $\cos \theta$

iii) $\tan \theta$

iv) $\csc \theta$

v) $\sec \theta$

vi) $\cot \theta$

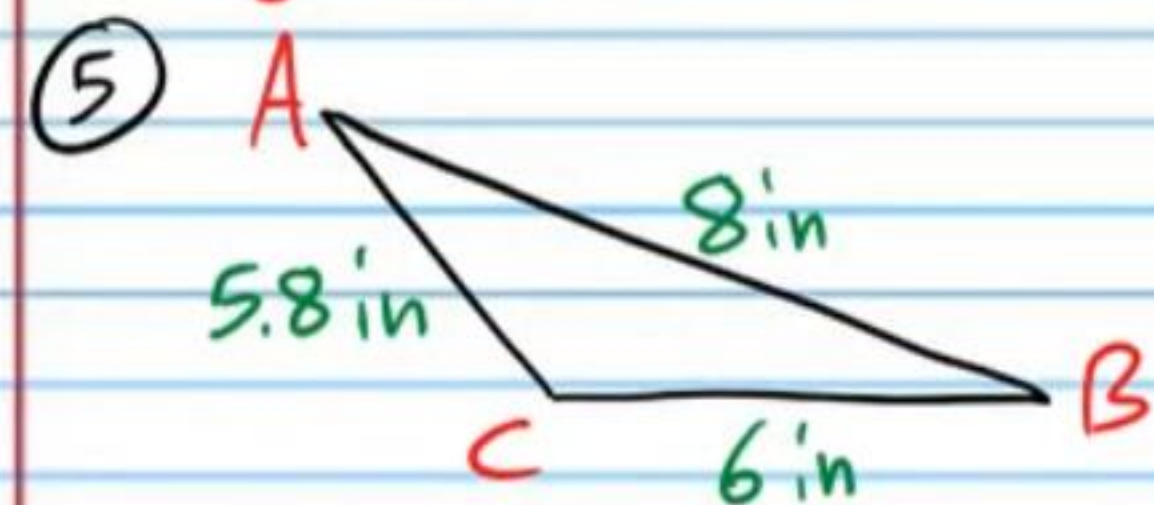
③ Solve if $0 \leq x \leq 360^\circ$.

$$\cot\left(\frac{x}{2} - 180^\circ\right) = 1$$

④ Solve if $0 < w < 2\pi$. Write your answer(s) in radian units.

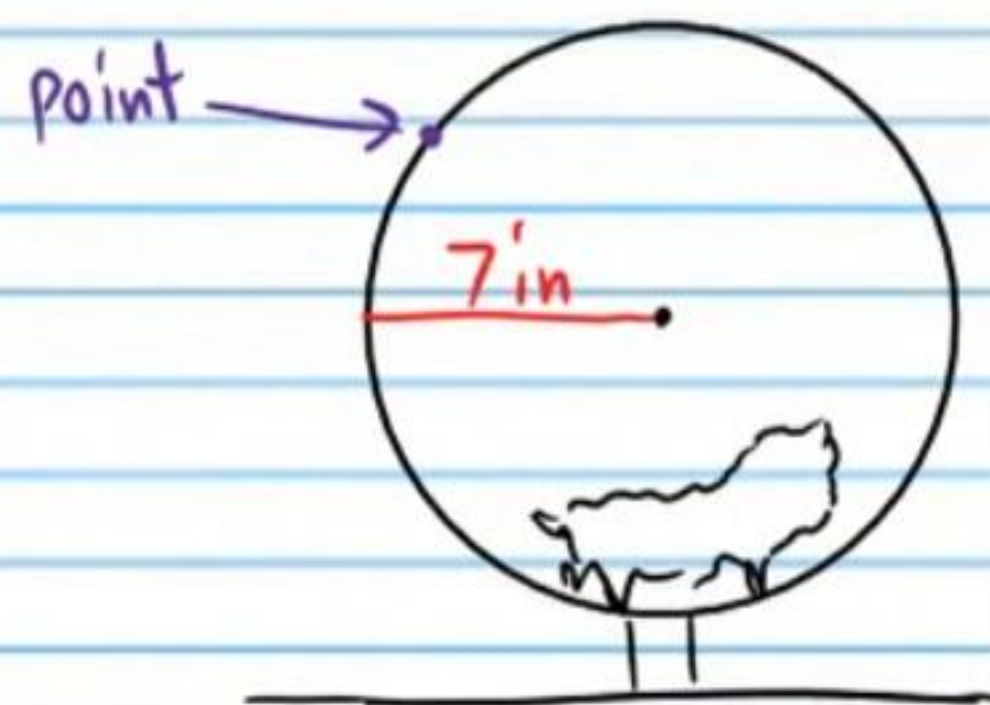
$$-5\tan w = 2\cos w + \sec w$$

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle. A triangle may not be possible for some problems.



⑥ $c = 12 \text{ km}$, $b = 8 \text{ km}$, $C = 103^\circ$

⑦ A hamster runs at a constant rate on a wheel. The wheel rotates at 240 rpm. If the radius of the wheel is 7 inches, how fast does a point on the wheel move in inches per second?



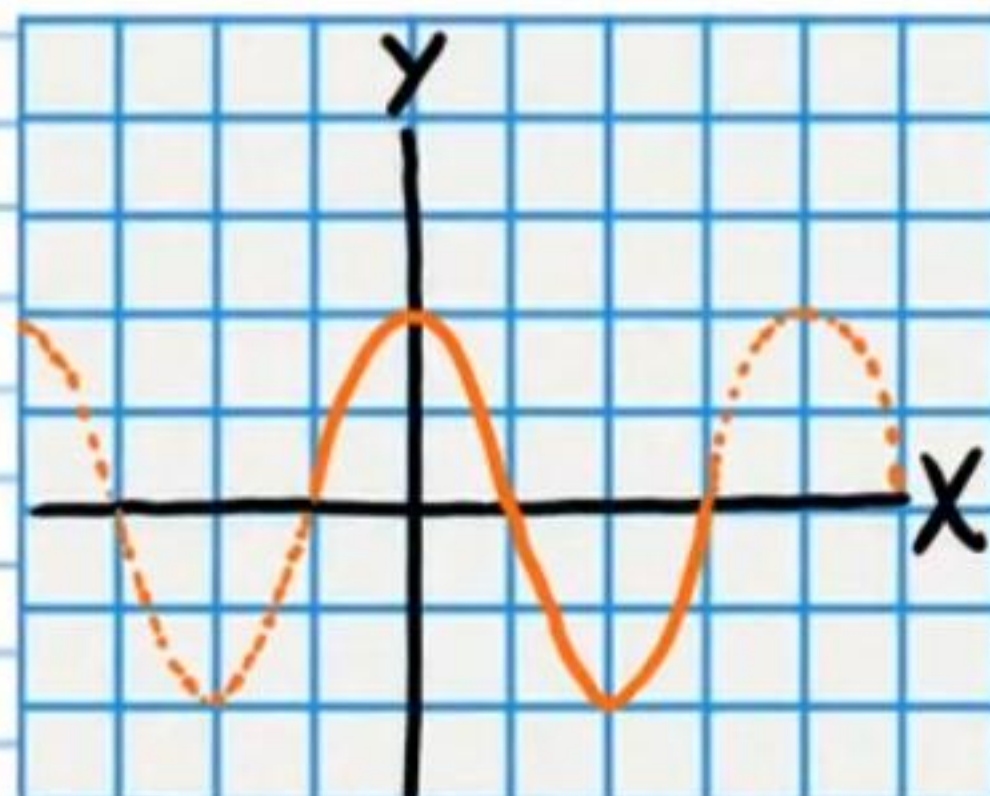
⑧ A Frisbee rotates at a constant angular velocity. It completes one rotation in 0.1 seconds. If the diameter of the Frisbee is 28 centimeters, how fast does the outer edge move in centimeters per second?

⑨ Write an equation corresponding to the graph below. Use the sine function rather than the cosine function. Each horizontal unit on the graph is 90° . Each vertical unit is one.

Midline Equation:

Amplitude:

Omega:



Phase Shift:

⑩ Graph the equation below. Write the period, phase shift, and the equation of the midline.

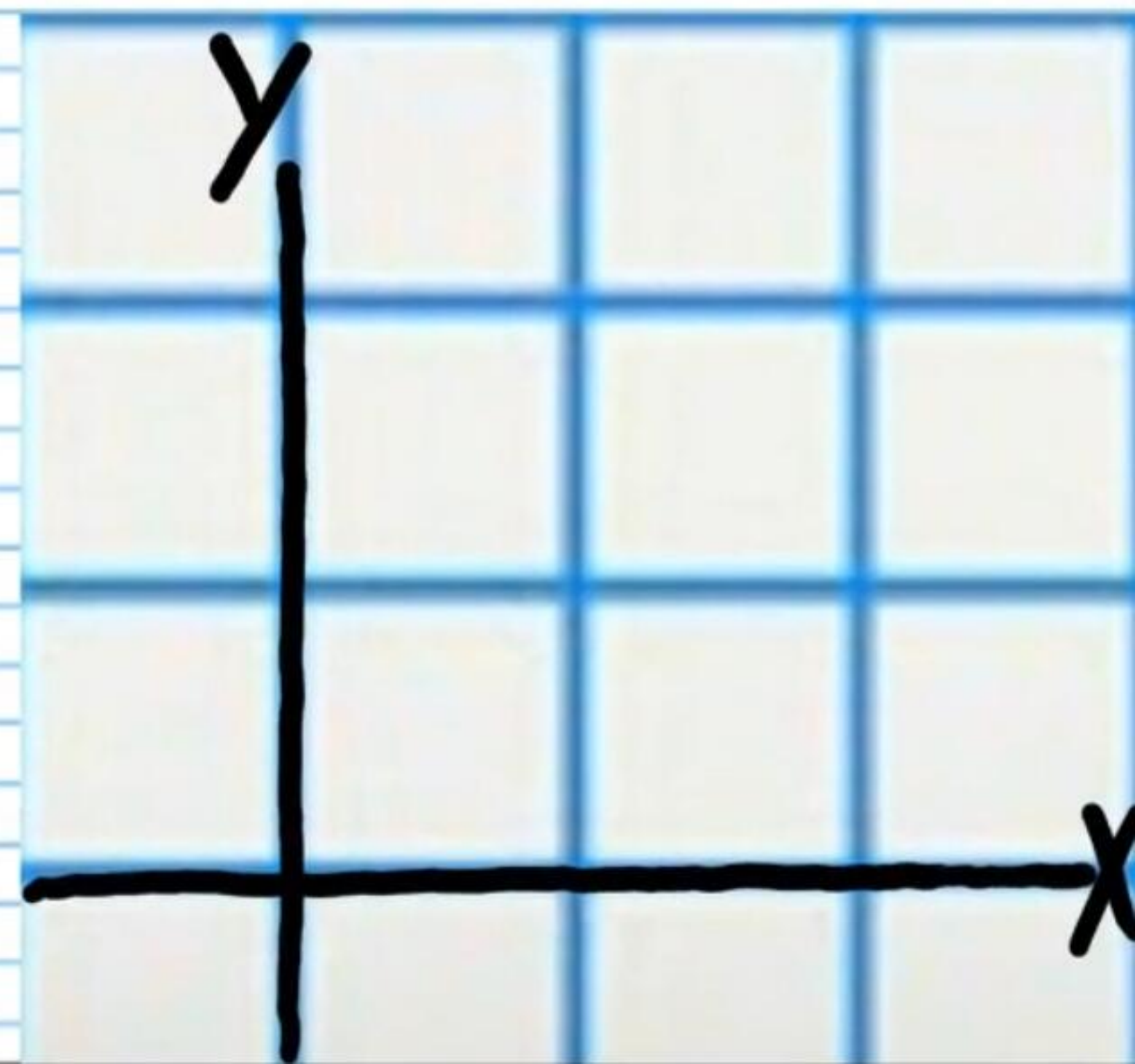
$$y = \tan(2x - 360^\circ) + 1$$

Midline Equation:

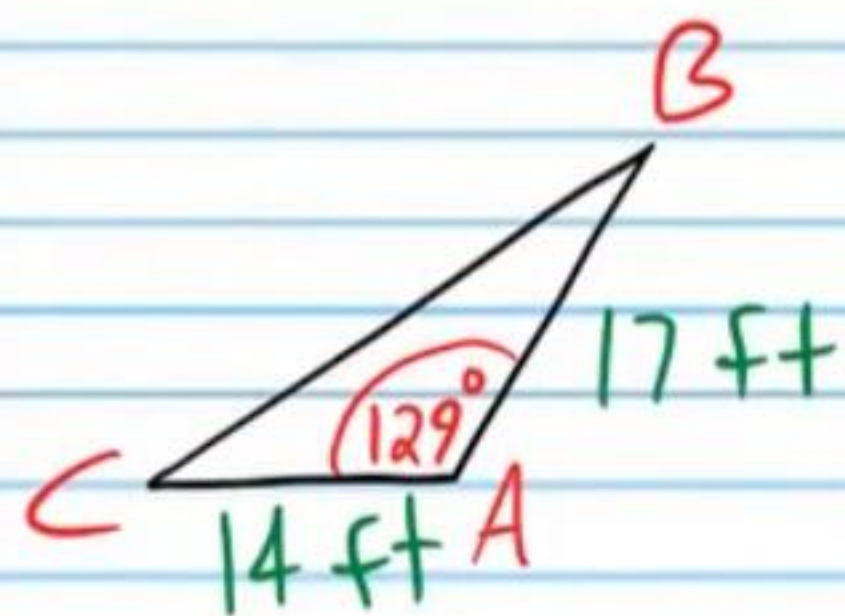
Period:

Phase Shift:

A: ~~*~~ Assume $A=1$.



⑪ Find the area of the triangle below. Round to the nearest tenth.



⑫ If θ is in standard position, and it terminates in QII ($90^\circ < \theta < 180^\circ$), find the value of each expression below if $\cos \theta = -\frac{2}{3}$.

i) $\sin(2\theta)$

ii) $\sin\left(\frac{\theta}{2}\right)$

⑬ Simplify the expression below.

$$\sin(12x)\cos(9x) + \sin(9x)\cos(12x)$$

Prove that the following equations are true by converting the left side to the right.

⑭ $\frac{\cos(2\theta)}{\sin^2\theta} = \cot^2\theta - 1$ ⑮ $\cos A \sin A \tan A \csc A = \sin A$

⑯ $\cot B + \tan B = \sec^2 B \cot B$

⑰ $\sin(90^\circ - W) + \cos(-W) = 2\cos W$

⑮ $\cos(6\beta) + \sin\beta \sin(5\beta) = \cos\beta \cos(5\beta)$

⑮ Solve if $0 \leq \theta < 2\pi$. Write your answer(s) in radian units.

$$\sin\theta + \sin(2\theta) = 3\sin\theta$$

②0 Solve if $0 < \theta < 2\pi$. Write your answer(s) in radian units.
Round to the nearest hundredth.

$$-4 \csc \theta + 9 = 0$$

②1 Find all values of W .

$$18 \cos W - 8 = 2 \cos W$$

Find the values of the following expressions **without a calculator.**

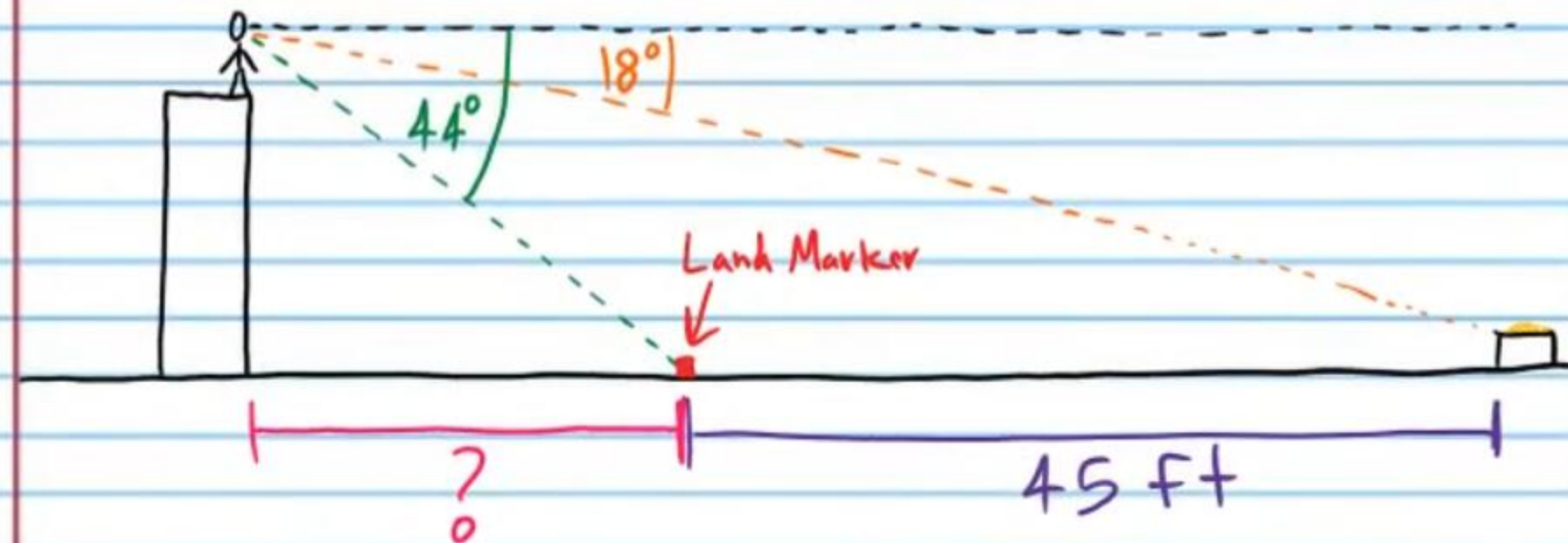
(22) $\sin(\cot^{-1} 0)$

(23) i) $\tan(\tan^{-1} \frac{1}{\sqrt{3}})$

ii) $\cos^{-1}(\cos(-120^\circ))$

24) $\sin^{-1}\left(\cos\left(\frac{\pi}{6}\right)\right)$

25) A man on a tower looks down at a treasure chest with an 18° angle of depression. He then looks down at a land marker with a 44° angle of depression. If the chest and the land marker are 45 feet apart, how far is the land marker from the base of the tower?



Practice Final Test (Version 2) Answers

① -45°

② i) $-\frac{5\sqrt{34}}{34}$

ii) $\frac{3\sqrt{34}}{34}$

iii) $-\frac{5}{3}$

iv) $-\frac{\sqrt{34}}{5}$

v) $\frac{\sqrt{34}}{3}$

vi) $-\frac{3}{5}$

③ 90°

④ $\frac{7\pi}{6}, \frac{11\pi}{6}$

⑤ $A \approx 48.38^\circ$
 $B \approx 46.27^\circ$
 $C \approx 85.35^\circ$

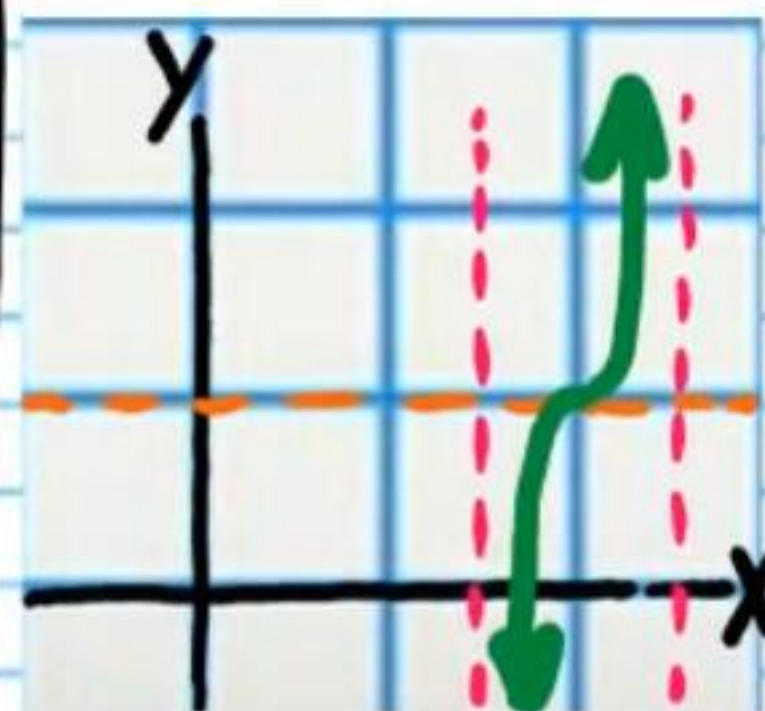
⑥ $a \approx 7.32 \text{ km}$
 $A \approx 36.49^\circ$
 $B \approx 40.51^\circ$

⑦ $56\pi \text{ in/s}$ or 175.9 in/s

⑧ $280\pi \text{ cm/s}$ or 879.6 cm/s

⑨ $y = 2\sin(x + 90^\circ)$

⑩ $y = 1$
 $P = 90^\circ$
 $PS = 180^\circ$



⑪ $92.5 f + 2$

⑫ i) $-\frac{4\sqrt{5}}{9}$ ii) $\frac{\sqrt{30}}{6}$

⑬ $\sin(21x)$

⑭ $\frac{\cos(2\theta)}{\sin^2\theta} = \cot^2\theta - 1$

$\frac{\cos^2\theta - \sin^2\theta}{\sin^2\theta} =$

$\frac{\cos^2\theta}{\sin^2\theta} - \frac{\sin^2\theta}{\sin^2\theta} =$

$\cot^2\theta - 1 =$



⑮ $\cos A \sin A \tan A \csc A = \sin A$

$\cos A \sin A \frac{\sin A}{\cos A} \frac{1}{\sin A} =$

$\sin A =$



$$(16) \cot B + \tan B = \sec^2 B \cot B$$

$$\frac{1}{\tan B} + \tan B =$$

$$\frac{1}{\tan B} + \frac{\tan^2 B}{\tan B} =$$

$$\frac{1 + \tan^2 B}{\tan B} =$$

$$\frac{\sec^2 B}{\tan B} =$$

$$\sec^2 B \frac{1}{\tan B} =$$

$$\sec^2 B \cot B =$$



$$(17) \sin(90^\circ - W) + \cos(-W) = 2 \cos W$$

$$\cos W + \cos W =$$

$$2 \cos W =$$



$$(18) \cos(6\beta) + \sin \beta \sin(5\beta) = \cos \beta \cos(5\beta)$$

$$\cos(\beta + 5\beta) + \sin \beta \sin(5\beta) =$$

$$\cos \beta \cos(5\beta) - \sin \beta \sin(5\beta) + \sin \beta \sin(5\beta) =$$

$$\cos \beta \cos(5\beta) =$$



$$(19) \boxed{0, \pi}$$

$$(20) \boxed{0.46, 2.68}$$

$$(21) \boxed{W = 60 + 360n, n \text{ is any integer.}}$$

$$\boxed{W = 300 + 360n, n \text{ is any integer.}}$$

$$(22) \boxed{1}$$

$$(23) \text{ i) } \boxed{\frac{1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}}{3}} \quad \text{ii) } \boxed{120^\circ}$$

$$(24) \boxed{\frac{\pi}{3}}$$

$$(25) \boxed{22.818 \text{ ft}}$$